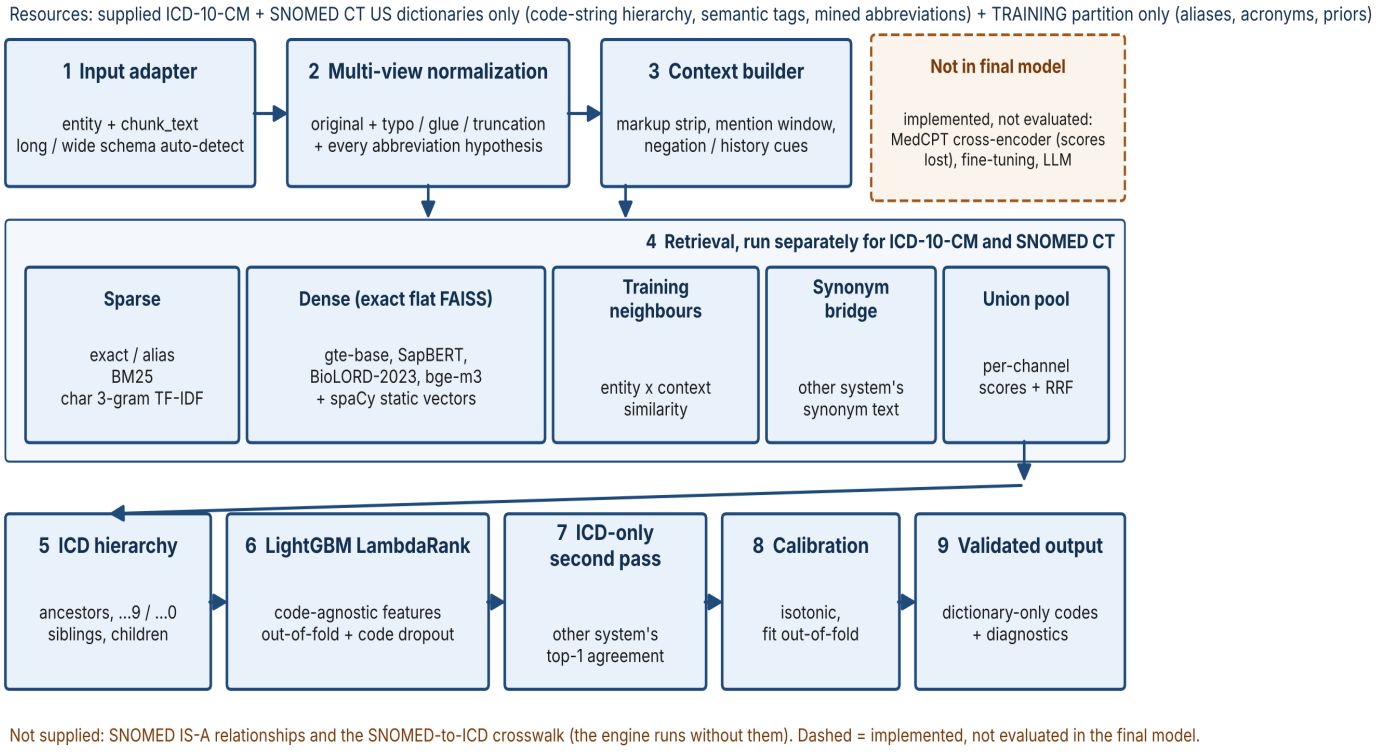
TermMap: mapping clinical entities to ICD-10-CM and SNOMED CT

HiLabs hackathon documentation (results status: **interim**; PENDING = to be filled from the final run). Demo video: [DEMO VIDEO LINK]. Repository: PENDING. No proprietary rows appear in this document; examples are from the problem statement or labelled synthetic.

1. Approach

Interpretation. For each extracted Problem entity (diagnosis, condition, sign or symptom) and its chunk text, return exactly one ICD-10-CM code and one SNOMED CT US concept, both present in the supplied dictionaries. Procedures, medications, labs and NER are out of scope, so only *DiagnosisCode* training rows are used. **Data facts.** 2,305 ICD and 2,700 SNOMED diagnosis rows; 2,882 entity+chunk pairs, 2,123 with both labels (per-system label masks everywhere). 72% of distinct SNOMED codes and about 54% of distinct ICD codes appear exactly once, and every entity string has exactly one code per system, so memorisation cannot reach the target and context cannot be learned from label conflicts. Malformed gold L6.0 (2 rows), reported, never remapped. Dictionaries: 159,306 ICD rows; 984,585 SNOMED descriptions for 382,124 concepts (exported from Term Map's gte_base_fp32 index). **Two independent systems.** ICD and SNOMED differ in granularity, so each is retrieved and ranked on its own; the other system is used only as text evidence, never as a crosswalk. **Protocol.** Group split by entity and chunk; validation for model selection; a code-disjoint fold (20% of codes removed from every training resource) for the long tail; a final test split scored once; bootstrap 95% CIs.

2. Architecture



Not supplied: SNOMED IS-A relationships and the SNOMED-to-ICD crosswalk (the engine runs without them). Dashed = implemented, not evaluated in the final model.

3. Design decisions

Hybrid sparse + dense retrieval. Before the dense encoders the correct code was in the top 200 only about 0.83 of the time on validation. The fused union of sparse channels and four encoders reaches @100 ICD 0.937 / SNOMED 0.966 and @200 ICD 0.964 / SNOMED 0.974 (code-disjoint @200: 0.937 / 0.954). Recall is a shortlist ceiling, not accuracy. **Code-agnostic LightGBM reranker.** A single encoder's top-1 is only ICD 0.20 to 0.33 and SNOMED 0.43 to 0.47, so ranking is the weak step; LambdaRank combines all channels' scores and ranks, string, context, qualifier and hierarchy features, none of which encode code identity, so unseen codes compete equally. **Code dropout.** Training tables are out-of-fold and a share of queries lose every trace of their gold code, so the model learns to decide without memory. **Learned specificity.** Ancestors, ...9/...0 siblings and children of top ICD candidates join the pool; depth, leaf and 'candidate adds/drops a modifier' features let the reranker pick the level. **ICD-only second pass.** Using the other system's top-1 text raised ICD from 0.608 to 0.618 on validation; it did not help SNOMED, so SNOMED keeps stage 1.

split	ICD-10-CM [95% CI]	SNOMED CT [95% CI]	headline (lower)	seeds / n (ICD, SNOMED)
Validation (codes seen)	0.618 PENDING	0.565 PENDING	0.565	3 / PENDING, PENDING
Code-disjoint (codes never seen)	0.436 [0.389, 0.484]	0.535 [0.487, 0.581]	0.436	2 / 399, 458
Final test (scored once)	PENDING PENDING	PENDING PENDING	PENDING	PENDING / PENDING, PENDING
Earlier CPU-only run (no encoders)	val 0.567; CD 0.328	val 0.504; CD 0.454		not like-for-like on CD rows

validation ablation	ICD	SNOMED
full model (stage 1 + ICD second pass)	0.618	0.565
stage 1 only (no second pass)	0.608	0.565
no dense encoders	0.570	0.528
no ICD hierarchy features	0.577	0.565
no training neighbours	0.595	0.559
no context features	0.607	0.567
no RRF fusion feature	0.612	0.563
no qualifier features	0.612	0.559
no noise-augmented rows	0.606	0.568
no learned priors	0.617	0.571

Target 95% on both systems: **not reached**; validation headline 0.565 is 0.385 below it. **Judging criteria.** Accuracy (50%): table above. Long tail (20%): code-disjoint row, code dropout, code-agnostic features. Engineering (15%): one command, offline, deterministic, tests, hash-keyed resumable caches. Error analysis (15%): section 4.

4. Error analysis

challenge failure mode	technique	measured evidence
1 nearest string is the wrong concept (hypertension vs pulmonary hypertension I27.20)	reranker over string + context + hierarchy features	counts **PENDING**
2 abbreviation collision (MS / RA / PID)	all expansions kept as hypotheses; context features	ICD abbreviation slice 0.600; MS / RA / PID ICD hard cases fail
3 specificity (diabetic neuropathy E11.40 vs diabetes E11.9)	hierarchy expansion + learned level	no-hierarchy ablation ICD 0.577; E13.40 error
4 ICD / SNOMED granularity mismatch	independent systems; ICD-only second pass	second pass ICD 0.608 -> 0.618
5 long tail / unseen codes	code-agnostic features, code dropout	ICD unseen 0.420; code-disjoint headline 0.436
6 OCR / truncation ('chronickidney disease stag 3')	glue split, truncation completion, typo views	hard case **PENDING** (final model)
7 terminology version drift	hash-keyed rebuild, code-agnostic model, drift test	modified-dictionary test (see tests)

Retrieval miss vs rank miss: **PENDING**; with union recall @200 near 0.96 to 0.97, most remaining errors are ranking errors. **ICD accuracy by training frequency of the gold code** (validation): unseen 0.420, 1 time 0.680, 2-5 times 0.680, 6+ 0.700. SNOMED by frequency: **PENDING**. **Weak ICD chapters:** F 0.36 (n=11); M, E and I about 0.60 to 0.64. **Calibration:** ECE ICD **PENDING**, SNOMED **PENDING**; accuracy at confidence >= 0.7: **PENDING**. **Hard cases (statement or synthetic):** HTN (statement): statement-vs-label conflict: training maps 'hypertension' and 'htn (hypertension)' to SNOMED 38341003, statement shows 59621000; training labels win; diabetic neuropathy (statement): FAIL: ICD E13.40 instead of E11.40; MS with mitral-valve context (synthetic): FAIL on the ICD side; RA (synthetic, joint and atrium contexts): FAIL on the ICD side; PID (synthetic, pelvic and disc contexts): FAIL on the ICD side for at least one sense; chronickidney disease stag 3 (statement): **PENDING** (final model); chest pain (synthetic): **PENDING** (final model).

5. Innovations (implemented and measured)

Code-dropout out-of-fold reranking for unseen codes (code-disjoint headline 0.436). **Training-mined local conventions** via neighbours: removing them drops ICD to 0.595. **Hierarchy-aware specificity**: removing it drops ICD to 0.577. **Learned qualifier relevance**: removing qualifier features gives 0.612 / 0.559 (within noise). **Dictionary-text synonym bridge** and **dual-system agreement** (ICD-only second pass, +0.010 ICD; agreement also feeds the confidence / review flag). **Did nothing measurable:** learned priors (removing them: 0.617 / 0.571), RRF and noise-augmented rows (within noise).

6. Challenges and learnings

What did not work or was lost: priors, RRF and augmented rows stayed within noise; the MedCPT cross-encoder was trained but its scores were lost in a Kaggle session reset, so the final model runs without it (implemented, not evaluated); contrastive fine-tuning and LLM adjudication were not run; RAM (8 GB sandbox) and disk limits forced per-split processes and hash-keyed resumable caches; a statement-versus-label conflict (HTN) where training labels win. **Engineering:** CPU-only about 6 terms/s, median 0.340 s per term, peak RAM about 5.5 GB; GPU throughput **PENDING**; per-term model calls: none (no cross-encoder or LLM in the final model); determinism: identical output hashes across two runs, verified in one environment only; tests **PENDING**. **Next:** re-run and measure the cross-encoder and fine-tuning; SNOMED IS-A and the crosswalk as auxiliary features; LLM adjudication only on low-confidence rows; larger candidate pool.

Open questions for the organisers
Evaluation schema (long vs wide) | partial credit? | multiple valid codes / statement-vs-label conflicts | annotator agreement | per-category floors | scoring of rows with one label | approval for running on a hosted notebook (Kaggle)

External resources: thenlper/gte-base (c078288308d8dee004ab72c6191778064285ec0c, MIT); cambridgeltl/SapBERT-from-PubMedBERT-fulltext (090663c3ae57bf35ffe4d0d468a2a88d03051a4d, Apache-2.0); FremyCompany/BioLORD-2023 (167aab527b238a50ca65224e6319215d2ff4fc9f, other: verify on model card); BAAI/bge-m3 (5617a9f61b028005a4858fdac845db406aefb181, MIT); ncbi/MedCPT-Cross-Encoder (implemented, not evaluated) (71caf65d4927987813984f54c284405a13fcca49, other: verify on model card); spaCy en_core_web_lg 3.8.0 (release wheel, verify on model card); LightGBM, scikit-learn, RapidFuzz, FAISS (exact flat index) (pinned in requirements.txt, MIT / BSD-3 / MIT / MIT). Supplied ICD-10-CM and SNOMED CT US exports (SNOMED licence applies). No API calls; GPU stages ran in a private Kaggle notebook.